

#2.5

1/9 건너뛰

2.19) Show that if $P(A|B) = P(A)$, then $P(B|A) = P(B)$ (This proves Lemma 2.2.12)• If $P(A|B) = P(A)$, A and B are independent events.By the multiplication rule, $P(A \cap B) = P(A|B)P(B)$, and by substitution, when A and B are independent, $P(A \cap B) = P(A)P(B)$.Considering $P(B|A) = \frac{P(A \cap B)}{P(A)}$, and again using substitution,

$$P(B|A) = \frac{P(A)P(B)}{P(A)} = P(B).$$

2.20 Use definition of independent events to show that:

a) if A and B are independent, then A and B^c are also independent▷ Since $P(A \cap B) = P(A)P(B)$,▷ Then $P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B^c)$

Q5) Exercise 2.25

a) $P(M \text{ or } Th) = P(M) + P(Th)$

$$= 0.20 + 0.27 = \boxed{0.47}$$

b) D = defective

$$P(D) = (0.2)(0.02) + (0.25)(0.015) + (0.28)(0.01) + (0.27)(0.03)$$

$$= 0.004 + 0.00375 + 0.0028 + 0.0081 = 0.01865$$

$$P(M \text{ or } Th | D) = \frac{(0.2)(0.02) + (0.27)(0.03)}{0.01865} = \frac{0.0121}{0.01865} = \boxed{0.6488}$$

c) π = function properly

$$P(\pi) = 1 - P(D) = 1 - 0.01865 = 0.98135$$

For Monday or Thursday and function properly

$$(0.20)(0.98) + (0.27)(0.97) = 0.196 + 0.2619 = 0.4579$$

$$\text{so, } P(M \text{ or } Th | \pi) = \frac{0.4579}{0.98135} \approx \boxed{0.466}$$

2.31

(a) ~~$p(\text{reject} | 10 \text{ defective})$~~

$$p(\text{all 4 is good}) = \frac{90}{100} \cdot \frac{89}{99} \cdot \frac{88}{98} \cdot \frac{87}{97}$$

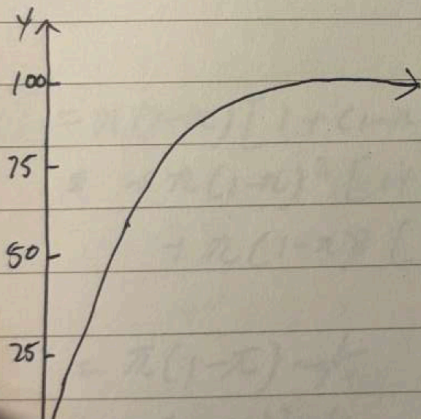
$$p(\text{reject}) = 1 - p(\text{all 4 is good}) = 0.3484$$

Even if the detection rate is 410%,
the rejection rate could still be 34.84%.

$$b) y = 100 \left(1 - \frac{(100-x)(99-x)(98-x)(97-x)}{100 \cdot 99 \cdot 98 \cdot 97} \right)$$

y = probability of rejection if draws 4 time out of 100 production

x = number of defection in 100 production



2.37 8 blue socks 4 brown 5 blue

how many 3 combination

$$\binom{17}{3} = 680$$

(a) a matching pair means not a color of each

$$p(\text{a color of each}) = \frac{\binom{8}{1} \cdot \binom{4}{1} \cdot \binom{5}{1}}{\binom{17}{3}} = \frac{8 \cdot 4 \cdot 5}{680} = \frac{4}{17} = 0.2353$$

$$p(\text{match}) = 1 - \frac{4}{17} = \frac{13}{17} = 0.7647$$

(b) $p(\text{a pair of black socks})$

exactly 2 black and 1 other $\binom{8}{2} \cdot \binom{9}{1} = 252$

all 3 are black = $\binom{8}{3}$

$$p(\text{two pair of black}) = \frac{\binom{8}{2} \cdot \binom{9}{1} + \binom{8}{3}}{\binom{17}{3}} = \frac{252 + 56}{680}$$

$$= \frac{77}{170} \approx 0.4529$$

2.43

x	1	2	3	4
$P(X=x)$	$\frac{4}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{1}{10}$

$$1 \rightarrow \frac{2}{5}$$

$$2 \rightarrow \frac{3}{5} \cdot \frac{2}{4} = \frac{6}{20} = \frac{3}{10}$$

$$3 \rightarrow \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3} = \frac{12}{60} = \frac{2}{10}$$

$$4 \rightarrow \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} \cdot \frac{2}{2} = \frac{12}{120} = \frac{1}{10}$$